MODULE -I

Numerical Solutions of Algebraic and Transcendental Equations

Introduction :

An expression of the form $(x) = a_0x^n + a_1x^{n^{-1}} + \cdots + a_{n-1}x + a_n, 0 \neq 0$ is called a polynomial of degree 'n' and the polynomial $f(x) = 0$ is called an algebraic equation of nth degree. If (x) contains trigonometric, logarithmic or exponential functions, then $(x)=0$ is called a transcendental equation. For example $x^2 + 2\sin x + e^x = 0$ is a transcendental equation.

If (x) is an algebraic polynomial of degree less than or equal to 4, direct methods for finding the roots of such equation are available. But if (x) is of higher degree or it involves transcendental functions, direct methods do not exist and we need to apply numerical methods to find the roots of the equation $f(x) = 0$.

Polynomial : An expression of the form $f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + a_3x^{n-3} + \ldots\ldots + a_{n-1}x + a_n$ is called a polynomial of 'x' of degree 'n' where $a_0, a_1, a_2, \ldots\ldots a_n$ are constants.

Algebraic equation : A polynomial in 'x' of degree 'n' when equated to zero *i.e.*, $f(x) = 0$ is called an algebraic equation.

Example : $x^4 - x - 10 = 0$

Transcendental equation: If the polynomial $f(x)$ involves the functions of the form such as trigonometric, logarithmic, exponential etc., then $f(x) = 0$ is called transcendental equation.

Example : $xe^x - 1 = 0$

Some useful results:

- ✓ If α is root of the equation $f(x)=0$, then $f(\alpha)=0$
- ✓ Every equation of nth degree has exactly n roots (real or imaginary)
- ✓ **Intermediate Value Theorem**: If $f(x)$ is a continuous function in a closed interval $[a,b]$ and $f(a)$ & $f(b)$ are having opposite signs, then the equation

$f(x)$=0 has at least one real root or odd number of roots between 'a' and 'b'.

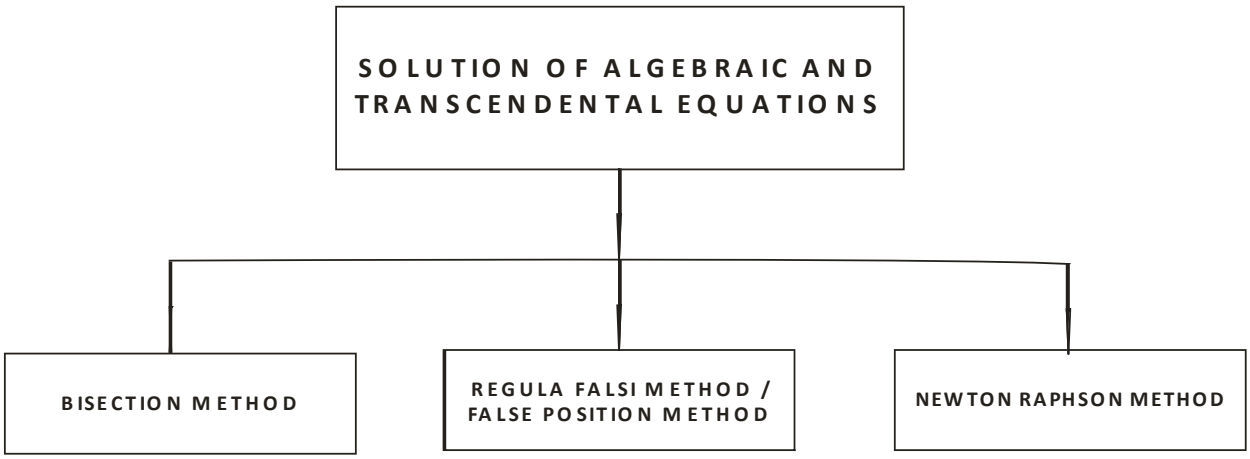
- ✓ If $f(x)$ is a continuous function in the closed interval $[a,b]$ and $f(a)$ & $f(b)$ are of same signs, then the equation $f(x)$ = 0 has no root or even number of roots between $a$and b.

Numerical methods to find roots of algebraic and transcendental equations :

Most numerical methods use iterative procedures to find an approximate root of an equation (x) = 0. They require an initial guess of the root as starting value and each subsequent iteration leads closer to the actual root.

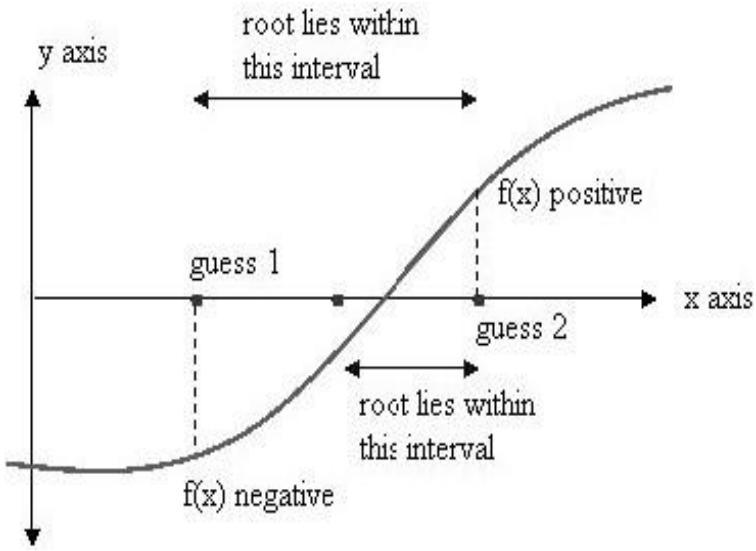
Order of convergence: For any iterative numerical method, each successive iteration gives an approximation that moves progressively closer to actual solution. This is known as convergence. Any numerical method is said have order of convergenceρ, ifρ is the largest positive number such that $|\epsilon_{n+1}| \leq k\ |\epsilon_n|^{\rho}$,where ϵ_n and ϵ_{n+1} are errors in

n^{t^h} and $(n+1)^h$ iterations, k is a finite positive constant.



Bisection Method(or Bolzano Method)

Bisection method is used to find an approximate root in an interval by repeatedly bisecting into sub intervals. It is a very simple and robust method but it is also relatively slow. Because of this it is often used to obtain a rough approximation to a solution which is then used as a starting point for more rapidly converging methods. This method is based on the intermediate value theorem for continuous functions.



Procedure :

Let $f(x)$ be a continuous function in the interval $[a,b]$, such that $f(a)$ and $f(b)$ are of opposite signs, i.e. $f(a)<0$ and $f(b)>0$.

Take the initial approximation given by $x_0 = \frac{a+b}{2}$

One of the three conditions arises for finding the 1st approximation x_1

- $f(x_0)=0$ We have a root at x_0
- If $f(x_0)<0$ then the root is lies between x_0 and b $\quad \therefore\ x_1 = \frac{x_0+b}{2}$
- If $f(x_0)>0$ then the root is lies between a and x_0 $\quad \therefore\ x_1 = \frac{a+x_0}{2}$
- Continue this process until root is found to be of desired accuracy.

Remarks : Convergence is not unidirectional as none of the end points is fixed. As a

result convergence of Bisection method is very slow.

Problem 1 : Apply bisection method to find a root of the equation $x^4 + 2x^3 - x - 1 = 0$

Solution: $(x) = x^4 + 2x^3 - x - 1$

Here $f(0) = 0^4 + 2.(0)^3 - 0 - 1 = -1 < 0$ and $f(1) = 1^4 + 2.(1)^3 - 1 - 1 = 1 > 0$

Also$(x)$is continuous in[0,1]

$\therefore$Root of $f(x)$ is lie between [0,1]

1st approximation: a=0 , b=1

$$x_0 = \frac{a+b}{2} = \frac{0+1}{2} = \frac{1}{2} = 0.5$$

$$f(0.5) = 0.5^4 + 2.(0.5)^3 - 0.5 - 1 = -1.1875 < 0$$

$\therefore$Root of $f(x)$ is lie between [0.5,1]

2nd approximation: a=0.5 , b=1

$$x_1 = \frac{a+b}{2} = \frac{0.5+1}{2} = \frac{1.5}{2} = 0.75$$

$$f(0.75) = 0.75^4 + 2.(0.75)^3 - 0.75 - 1 = -0.5898 < 0$$

$\therefore$Root of $f(x)$ is lie between [0.75,1]

3rd approximation: a=0.75 , b=1

$$\boldsymbol{x_2 = \frac{a+b}{2} = \frac{0.75+1}{2} = \frac{1.75}{2} = 0.875}$$

$$f(0.875) = 0.875^4 + 2.(0.875)^3 - 0.875 - 1 = 0.051 > 0$$

$\therefore$Root of $f(x)$ is lie between [0.75,0.875]

4th approximation: a=0.75 , b=0.875

$$\boldsymbol{x_3 = \frac{a+b}{2} = \frac{0.75+0.875}{2} = \frac{1.625}{2} = 0.8125}$$

$$f(0.8125) = 0.8125^4 + 2.(0.8125)^3 - 0.8125 - 1 = -0.3039 < 0$$

$\therefore$Root of $f(x)$ is lie between [0.8125,0.875]

5th approximation: a=0.8125 , b=0.875

$$\boldsymbol{x_4 = \frac{a+b}{2} = \frac{0.8125+0.875}{2} = \frac{1.6875}{2} = 0.8438}$$

$$f(0.8438) = 0.8438^4 + 2.(0.8438)^3 - 0.8438 - 1 = -0.1353 < 0$$

$\therefore$Root of $f(x)$ is lie between [0.8438,0.875]

6th approximation: a=0.8438, b=0.875

$$x_5 = \frac{a+b}{2} = \frac{0.8438+0.875}{2} = \frac{1.7188}{2} = \mathbf{0.8594}$$

$$f(0.8594) = 0.8594^4 + 2.(0.8594)^3 - 0.8594 - 1 = -0.0445 < 0$$

$\therefore$Root of $f(x)$ is lie between $[0.8594, 0.875]$

7th approximation: a=0.8594, b=0.875

$$x_6 = \frac{a+b}{2} = \frac{0.8594+0.875}{2} = \frac{1.7344}{2} = \mathbf{0.8672}$$

$$f(0.8672) = 0.8672^4 + 2.(0.8672)^3 - 0.8672 - 1 = 0.0027 > 0$$

$\therefore$Root of $f(x)$ is lie between $[0.8594, 0.8672]$

8th approximation: a=0.8594, b=0.8672

$$x_7 = \frac{a+b}{2} = \frac{0.8594+0.8672}{2} = \frac{1.7266}{2} = \mathbf{0.8633}$$

$$f(0.8633) = 0.8633^4 + 2.(0.8633)^3 - 0.8633 - 1 = -0.021 < 0$$

$\therefore$Root of $f(x)$ is lie between $[0.8633, 0.8672]$

9th approximation: a=0.8633, b=0.8672

$$x_8 = \frac{a+b}{2} = \frac{0.8633+0.8672}{2} = \frac{1.7305}{2} = \mathbf{0.8653}$$

$$f(0.8653) = 0.8653^4 + 2.(0.8653)^3 - 0.8653 - 1 = -0.0089 < 0$$

$\therefore$Root of $f(x)$ is lie between $[0.8653, 0.8672]$

10th approximation: a=0.8653, b=0.8672

$$x_9 = \frac{a+b}{2} = \frac{0.8653+0.8672}{2} = \frac{1.7325}{2} = \mathbf{0.8663}$$

$$f(0.8663) = 0.8663^4 + 2.(0.8663)^3 - 0.8663 - 1 = -0.0028 < 0$$

$\therefore$Root of $f(x)$ is lie between $[0.8663, 0.8672]$

11th approximation: a=0.8663, b=0.8672

$$x_{10} = \frac{a+b}{2} = \frac{0.8663+0.8672}{2} = \frac{1.7335}{2} = \mathbf{0.8668}$$

$$f(0.8668) = 0.8668^4 + 2.(0.8668)^3 - 0.8668 - 1 = 0.0002 > 0$$

$\therefore$Root of $f(x)$ is lie between $[0.8663, 0.8668]$

12th approximation: a=0.8663, b=0.8668

$$x_{11} = \frac{a+b}{2} = \frac{0.8663+0.8668}{2} = \frac{1.7331}{2} = \mathbf{0.8666}$$

$$f(0.8666) = 0.8666^4 + 2.(0.8666)^3 - 0.8666 - 1 = -0.001 < 0$$

$\therefore$Root of $f(x)$ is lie between $[0.8666, 0.8668]$

13th approximation: a=0.8666, b=0.8668

$$x_{12} = \frac{a+b}{2} = \frac{0.8666+0.8668}{2} = \frac{1.7334}{2} = 0.8667$$

$\therefore$Root of $f(x)$ is 0.8667

Problem 2 : Apply bisection method to find a root of the equation $x\log_{10} x = 1.2$ upto 5 approximations.

Solution: $f(x) = x\log_{10} x - 1.2$

Put $x=1$ then $f(1) = 1.\log(1) - 1.2 = -1.2$

Put $x=2$ then $f(2) = 1.\log(2) - 1.2 = -0.5979 < 0$

Put $x=3$ then $f(3) = 1.\log(3) - 1.2 = 0.2313 > 0$

$\therefore$ Root of $f(x)$ is lie between 2 and 3

1st approximation: a=2 , b=3

$$x_1 = \frac{a+b}{2} = \frac{2+3}{2} = \frac{5}{2} = 2.5$$

$$f(2.5) = (2.5)\log(2.5) - 1.2 = -0.2051$$

$\therefore$Root of $f(x)$ is lie between 2.5 and 3

2nd approximation: a=2.5 , b=3

$$x_2 = \frac{a+b}{2} = \frac{2.5+3}{2} = \frac{5.5}{2} = 2.75$$

$$f(2.75) = (2.75)\log(2.75) - 1.2 = 0.0082$$

$\therefore$Root of $f(x)$ is lie between 2.5 and 2.75

3rd approximation: a=2.5 , b=2.75

$$x_3 = \frac{a+b}{2} = \frac{2.5+2.75}{2} = \frac{5.25}{2} = 2.625$$

$$f(2.625) = (2.625)\log(2.625) - 1.2 = -0.0998$$

$\therefore$Root of $f(x)$ is lie between 2.625 and 2.75

4th approximation: a=2.625 , b=2.75

$$x_4 = \frac{a+b}{2} = \frac{2.625+2.75}{2} = \frac{5.375}{2} = 2.6875$$

$$f(2.6875) = (2.6875)\log(2.6875) - 1.2 = -0.0461$$

$\therefore$Root of $f(x)$ is lie between 2.6875 and 2.75

5th approximation: a=2.6875 , b=2.75

$$x_5 = \frac{a+b}{2} = \frac{2.6875+2.75}{2} = \frac{5.4375}{2} = 2.7188$$

$$f(2.7188) = (2.7188)\log(2.7188) - 1.2 = -0.0461$$

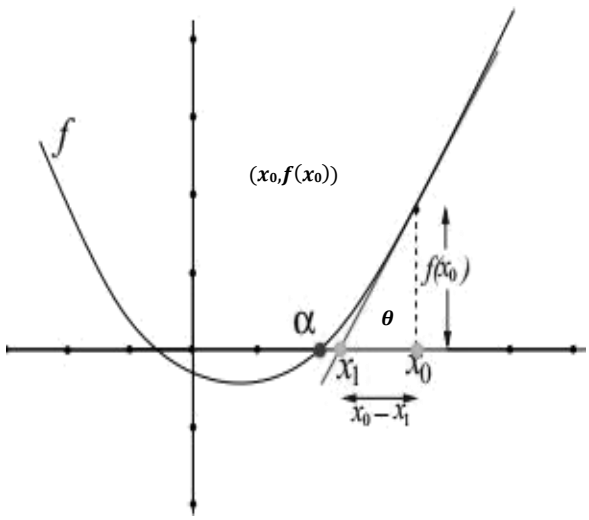
$\therefore$Root of $f(x)$ is lie between 2.7188 and 2.75

$\therefore$ Root of $f(x)$ is 2.7188 (Approximately)

Newton- Raphson Method (Geometrical Interpretation)

Newton-Raphson method named after Isaac Newton and Joseph Raphson is a powerful technique for solving equations numerically. The Newton-Raphson method in one variable is implemented as follows:

Let α be an exact root and x_0 be the initial approximate root of the equation $f(x)=0$.First approximation $x_1$is taken by drawing a tangent to curve$y$=(x) at the point (x_0, y_0). If θ is the angle which tangent through the point $(x_0, f(y_0))$ makes with x- axis, then slope of the tangent is given by:



$$Tan\theta = \frac{f(x_0)}{x_0 - x_1} = f'(x_0)$$

$$\frac{f(x_0)}{f'(x_0)} = x_0 - x_1$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

Problem 1 : Use Newton-Raphson method to find a root of the equation $x^3-5x+3=0$ Correct to four decimal places.

Solution :

Let the given algebraic equation be $f(x) = x^3 - 5x + 3$

Put $x = 1$ then $f(1) = 1^3 - 5(1) + 3 = -1$

Put $x = 2$ then $f(2) = 2^3 - 5(2) + 3 = 1$

$\therefore$ Root of $f(x)$ is lie between 1 and 2

Let the initial approximation be $x_0 = \frac{a+b}{2} = \frac{1+2}{2} = 1.5$

$$f'(x) = \frac{d}{dx}x^3 - 5\frac{d}{dx}x + \frac{d}{dx}3 = 3x^2 - 5(1) + 0 = 3x^2 - 5$$

1st Iteration : $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$

$$x_1 = x_0 - \frac{x_0^3 - 5x_0 + 3}{3x_0^2 - 5}$$

$$x_1 = 1.5 - \frac{(1.5)^3 - 5(1.5) + 3}{3(1.5)^2 - 5} = 2.1429$$

2nd Iteration : $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$

$$x_2 = x_1 - \frac{x_1^3 - 5x_1 + 3}{3x_1^2 - 5}$$

$$x_2 = 2.1429 - \frac{(2.1429)^3 - 5(2.1429) + 3}{3(2.1429)^2 - 5} = 1.9007$$

3rd Iteration : $x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$

$$x_3 = x_2 - \frac{x_2^3 - 5x_2 + 3}{3x_2^2 - 5}$$

$$x_3 = 1.9007 - \frac{(1.9007)^3 - 5(1.9007) + 3}{3(1.9007)^2 - 5} = 1.8385$$

4th Iteration : $x_4 = x_3 - \frac{f(x_3)}{f'(x_3)}$

$$x_4 = x_3 - \frac{x_3^3 - 5x_3 + 3}{3x_3^2 - 5}$$

$$x_4 = 1.8385 - \frac{(1.8385)^3 - 5(1.8385) + 3}{3(1.8385)^2 - 5} = 1.8343$$

5th Iteration : $x_5 = x_3 - \frac{f(x_4)}{f'(x_4)}$

$$x_5 = x_4 - \frac{x_4^3 - 5x_4 + 3}{3x_4^2 - 5}$$

$$x_5 = 1.8343 - \frac{(1.8343)^3 - 5(1.8343) + 3}{3(1.8343)^2 - 5} = 1.8342$$

6th Iteration : $x_6 = x_5 - \frac{f(x_5)}{f'(x_5)}$

$$x_6 = x_5 - \frac{x_5^3 - 5x_5 + 3}{3x_5^2 - 5}$$

$$x_6 = 1.8342 - \frac{(1.8342)^3 - 5(1.8342) + 3}{3(1.8342)^2 - 5} = 1.8342$$

$\therefore$ Root of $f(x)$ is 1.8342

Problem 2 : Use Newton-Raphson method to find a root of the equation $xe^x - 1 = 0$ Correct to four decimal places.

Solution :

Let the given algebraic equation be $f(x) = xe^x - 1$

Put $x = 0$ then $f(0) = 0.e^0 - 1 = -1$

Put $x = 1$ then $f(1) = 1.e^1 - 1 = 1.7182$

$\therefore$ Root of $f(x)$ is lie between 0 and 1

Let the initial approximation be $x_0 = \frac{a+b}{2} = \frac{0+1}{2} = 0.5$

$$f'(x) = \frac{d}{dx} xe^x - \frac{d}{dx} 1$$

$$f'(x)=\left(\frac{d}{dx}x\right)e^{x}+x\left(\frac{d}{dx}e^{x}\right)-\frac{d}{dx}1$$

$$f'(x)=1.e^{x}+x\left(e^{x}\right)-0$$

$$f'(x)=e^{x}(x+1)$$

1st Iteration : $x_1 = x_0 - \dfrac{f(x_0)}{f'(x_0)}$

$$x_1 = x_0 - \frac{x_0 e^{x_0} - 1}{e^{x_0}(x_0+1)}$$

$$x_1 = 0.5 - \frac{0.5\ e^{0.5} - 1}{e^{0.5}(0.5+1)} = 0.5710$$

2nd Iteration : $x_2 = x_1 - \dfrac{f(x_1)}{f'(x_1)}$

$$x_2 = x_1 - \frac{x_1 e^{x_1} - 1}{e^{x_1}(x_1+1)}$$

$$x_2 = 0.5710 - \frac{0.5710\ e^{0.5710} - 1}{e^{0.5710}(0.5710+1)} = 0.5672$$

3rd Iteration : $x_3 = x_2 - \dfrac{f(x_2)}{f'(x_2)}$

$$x_3 = x_2 - \frac{x_2 e^{x_2} - 1}{e^{x_2}(x_2+1)}$$

$$x_3 = 0.5672 - \frac{0.5672\ e^{0.5672} - 1}{e^{0.5672}(0.5672+1)} = 0.5671$$

4th Iteration : $x_4 = x_3 - \dfrac{f(x_3)}{f'(x_3)}$

$$x_4 = x_3 - \frac{x_3 e^{x_3} - 1}{e^{x_3}(x_3+1)}$$

$$x_4 = 0.5671 - \frac{0.5671\ e^{0.5671} - 1}{e^{0.5671}(0.5671+1)} = 0.5671$$

$\therefore$ Root of $f(x)$ is 1.8342

FINITE DIFFERENCES

Introduction :

While solving the problems related to mathematics, we use the terms addition, subtraction, multiplication and division respectively. These terms are denoted with the symbols $+,-,\times$ *and* $\div$ respectively. Similarly the important operators that can be used for solving numerical problem are Δ, ∇ *and* E respectively.

Forward Difference Operator Δ :

Let 'x' be an independent variable which takes the values of equal interval $x_0, x_0+h, x_0+2h, x_0+3h, \ldots\ldots x_0+nh$

Let $y=f(x)$ be the function of independent variables, then the values of the dependent variable y are $f(x_0), f(x_0+h), f(x_0+2h), f(x_0+3h) \ldots\ldots\ldots f(x_0+nh)$ respectively.

If the values of the independent variable lies in equal intervals, then the operator that can be used to find the finite forward differences is called Forward difference operator Δ

That is, if the values of independent variable lies in equal interval 'h' then $\boxed{\Delta f(x)=f(x+h)-f(x)}$.

Second order finite difference of $f(x)$ is

$$\Delta^2 f(x)=\Delta(\Delta f(x))=\Delta[f(x+h)-f(x)]$$
$$=\Delta f(x+h)-\Delta f(x)$$
$$=[f(x+2h)-f(x+h)]-[f(x+h)-f(x)]$$
$$\Delta^2 f(x)=f(x+2h)-2f(x+h)+f(x)$$
$$\boxed{\Delta^2 f(x)={}^2C_0\, f(x+2h)-{}^2C_1\, f(x+h)+{}^2C_2\, f(x)}$$

Third order finite difference of $f(x)$ is

$$\Delta^3 f(x)=\Delta(\Delta^2 f(x))=\Delta[f(x+2h)-2f(x+h)+f(x)]$$
$$=\Delta f(x+2h)-2\Delta f(x+h)+\Delta f(x)$$
$$=[f(x+3h)-f(x+2h)]-2[f(x+2h)-f(x+h)]+[f(x+h)-f(x)]$$
$$=f(x+3h)-f(x+2h)-2f(x+2h)+2f(x+h)+f(x+h)-f(x)$$
$$\Delta^3 f(x)=f(x+3h)-3f(x+2h)+3f(x+h)-f(x)$$
$$\boxed{\Delta^3 f(x)={}^3C_0\, f(x+3h)-{}^3C_1\, f(x+2h)+{}^3C_2\, f(x+h)-{}^3C_3\, f(x)}$$

Similarly, $\Delta^4 f(x)={}^4C_0\, f(x+4h)-{}^4C_1\, f(x+3h)+{}^4C_2\, f(x+2h)-{}^4C_3\, f(x+h)+{}^4C_4\, f(x)$

$$\Delta^n f(x)={}^nC_0\, f(x+nh)-{}^nC_1\, f(x+(n-1)h)+{}^nC_2\, f(x+(n-2)h)-{}^nC_3\, f(x+(n-3)h)+\ldots\ldots\ldots+(-1)^n\, {}^nC_n\, f(x)$$

Uses :

1. Operator Δ can be used to find the finite difference of $f(x)$ at any order.
2. It can be used to construct forward difference table.
3. It can be used to estimate missing values of $y=f(x)$
4. It can be used to determine the unknown polynomial relationship between the $f(x)$ values.

Shifting Operator E :

If the values of independent variable lies in equal intervals then the operator that can be used to find the functional value of $x+h$ by the known functional value of 'x' is called as shifting operator E.

The values of 'x' lies in arithmetic series with the internal of difference 'h' then

$$\mathrm{E}\, f(x) = f(x+h)$$

Similarly, $\mathrm{E}^2\, f(x) = \mathrm{E}[\mathrm{E}\, f(x)] = \mathrm{E}[\, f(x+h)] = f(x+2h)$

$$\mathrm{E}^3\, f(x) = f(x+3h)$$

$$\mathrm{E}^4\, f(x) = f(x+4h)$$

$$\vdots \qquad \vdots$$

$$\mathrm{E}^n\, f(x) = f(x+nh)$$

Uses :

1. The operator E can be used to find increments in $f(x)$ at different orders.
2. It can be used to derive different interpolation and central difference formula.
3. It can be used to determine some unknown polynomial relationship between x and $f(x)$ values.

Backward Difference Operator ∇ :

By the values of independent variable lies at equal intervals with the interval of difference 'h' then the operator used to find finite backward differences of $f(x)$ is called as backward difference operator '∇'.

If the arguments (observations) are in the arithmetic series with the common difference 'h' then the operator ∇ can be defined as

$$\nabla\, f(x+h) = f(x+h) - f(x)$$

Similarly, $\nabla\, f(x+2h) = f(x+2h) - f(x+h)$

$$\nabla\, f(x+3h) = f(x+3h) - f(x+2h)$$

$$\vdots \qquad \vdots$$

$$\nabla\, f(x+nh) = f(x+nh) - f(x+(n-1)h)$$

Similarly, second order backward differences are

$$\begin{aligned}\nabla^2\, f(x+h) &= \nabla(\nabla f(x+2h)) \\ &= \nabla[f(x+2h) - f(x+h)] \\ &= \nabla f(x+2h) - \nabla f(x+h) \\ &= [f(x+2h) - f(x+h)] - [f(x+h) - f(x)]\end{aligned}$$

$$\nabla^2\, f(x+h) = f(x+2h) - 2f(x+h) + f(x)$$

Uses :

1. The operator ∇ can be used to find finite backward differences of $f(x)$
2. It can be used to derive Newton's backward and Gauss Backward formula.
3. It can be used to determine some unknown polynomial relationship between $f(x)$ values.

Relationship between the Operators Δ and E :

The Operator Δ can be defined as $\Delta\, f(x) = f(x+h) - f(x) \rightarrow (1)$

The Operator E can be defined as $\mathrm{E}\, f(x) = f(x+h) \rightarrow (2)$

From Equation (1) and Equation (2)

$$\Delta\, f(x) = \mathrm{E}\, f(x) - f(x))$$

$$\Delta\, f(x) = f(x)[\mathrm{E} - 1]$$

$$\Delta = \mathrm{E} - 1$$

$$\boxed{E = \Delta + 1}$$

$$\Delta E\ f(x) = \Delta[E\ f(x)] = \Delta[\ f(x+h)]$$

$$= f(x+2h) - f(x+h)$$

$$= E\ f(x+h) - E\ f(x)$$

$$= E[f(x+h) - f(x)]$$

$$\Delta E\ f(x) = E\Delta\ f(x)$$

$$\boxed{\Delta E = E\Delta}$$

$\therefore \Delta$ *and* E satisfies Commutative property

Relationship between ∇ *and* E:

The operator ∇ can be defined as $\nabla\ f(x+h) = f(x+h) - f(x) \rightarrow (1)$

The operator E can be defined as $E\ f(x) = f(x+h) \rightarrow (2)$

From equation (1) and equation (2)

$$\nabla\ f(x+h) = E\ f(x) - f(x)$$

$$\nabla\ f(x+h) = f(x)[E-1]$$

$$\nabla E\ f(x) = f(x)[E-1]$$

$$\nabla E = [E-1]$$

$$E - \nabla E = 1$$

$$E[1-\nabla] = 1$$

$$\boxed{E = \frac{1}{1-\nabla}}$$

Relationship between the operators Δ, ∇, E:

The operator Δ can be defined as $\Delta\ f(x) = f(x+h) - f(x) \rightarrow (1)$

The operator ∇ can be defined as $\nabla\ f(x+h) = f(x+h) - f(x) \rightarrow (2)$

From equation (1) and equation (2) right hand side terms are equal

$$\Delta\ f(x) = \nabla\ f(x+h) \rightarrow (3)$$

We know that, the operator E can be defined as,

$$E\ f(x) = f(x+h) \rightarrow (4)$$

From Equation (3) and Equation (4)

$$\Delta\ f(x) = \nabla E\ f(x)$$

$$\Delta = \nabla E$$

$$\boxed{E = \frac{\Delta}{\nabla}}$$

Result 1 :

Statement : Show that the operators Δ and E obey communicative law.

Proof : (i) Generally, the operator Δ can be defined as $\Delta\ f(x) = f(x+h) - f(x)$

Now , if $f(x)$ is any function of 'x' and 'C' is a constant, then

$$\Delta C\ f(x) = C\ f(x+h) - C\ f(x)$$

$$\Delta C\ f(x) = C[f(x+h) - f(x)]$$

$$\Delta C\ f(x) = C\Delta\ f(x) \rightarrow (1)$$

(ii) The operator E can be defined as

$$\mathrm{E}\, f(x) = f(x+h)$$

Now , if $f(x)$ is any function of 'x' and 'C' is a constant, then

$$\mathrm{E}[C\, f(x)] = C\, f(x+h)$$

$$\mathrm{E}[C\, f(x)] = C\,[\mathrm{E} f(x)] \rightarrow (2)$$

From Equation (1) and Equation (2)

WE can say that, the operators Δ and E obeys commutative law.

Result 2 : Show that the operators Δ and E obey distributive law.

Proof : (i) The operator Δ can be defined as $\Delta\, f(x) = f(x+h) - f(x)$

If $f(x)$ and $g(x)$ are any two functions of 'x' then

$$\Delta[f(x)+g(x)] = [f(x+h)+g(x+h)] - [f(x)+g(x)]$$

$$= [f(x+h) - f(x)] + [g(x+h) - g(x)]$$

$$\Delta[f(x)+g(x)] = \Delta\, f(x) + \Delta\, g(x) \rightarrow (i)$$

(ii) The operator E can be defined as

$$\mathrm{E}\, f(x) = f(x+h)$$

If $f(x)$ and $g(x)$ are any two functions of 'x' then

$$\mathrm{E}[f(x)+g(x)] = f(x+h) + g(x+h)$$

$$\mathrm{E}[f(x)+g(x)] = \mathrm{E}\, f(x) + \mathrm{E}\, g(x) \rightarrow (ii)$$

From Equation (i) and (ii), we can say that,

The operators Δ and E obey distributive law.

Properties of Δ and E:

1. The operators Δ and E follows commutative property.

   i.e., $\Delta\, \mathrm{C}\, f(x) = C\Delta\, f(x)$

   $\mathrm{E}[C\, f(x)] = C[\mathrm{E}\, f(x)]$

2. The operators Δ and E follows distributive law

   i.e., $\Delta[f(x)+g(x)] = \Delta\, f(x) + \Delta\, g(x)$

   $\mathrm{E}[f(x)+g(x)] = \mathrm{E}\, f(x) + \mathrm{E}\, g(x)$

3. If $f(x)$, $g(x)$ are any two functions of 'x' and a, b are any two constants then the operators Δ and E are linear.

   i.e., $\Delta[a\, f(x) + b\, g(x)] = a\, \Delta\, f(x) + b\Delta\, g(x)$

   $\mathrm{E}[a\, f(x) + b\, g(x)] = a\, \mathrm{E}\, f(x) + b\, \mathrm{E}\, g(x)$

4. The operator Δ and E follows laws of indices

   i.e., $\Delta^m\, \Delta^n = \Delta^{m+n}$

   $\mathrm{E}^m\, \mathrm{E}^n = \mathrm{E}^{m+n}$

Result : If $f(x)$ is any function of 'x' and 'n' is any positive integer then

$$f(x_0 + nh) = \mathrm{E}^n\, f(x_0) = {}^nC_0\, f(x_0) + {}^nC_1\, \Delta\, f(x_0) + {}^nC_2\, \Delta^2\, f(x_0) + \ldots\ldots\ldots + {}^nC_n\, \Delta^n\, f(x_0)$$

Proof : We know that , by the relationship between the operators Δ *and* E

$$\mathrm{E} = 1 + \Delta$$

The operator E can be defined as $\mathrm{E}\, f(x) = f(x+h)$

Similarly, $\mathrm{E}\, f(x_0) = f(x_0 + h)$

$$f(x_0 + h) = \mathrm{E}\, f(x_0)$$

$$f(x_0 + h) = (1+\Delta)\, f(x_0)$$

$$f(x_0+h)=f(x_0)+\Delta\, f(x_0)$$

$$\boxed{f(x_0+h)={}^1C_0\ f(x_0)+{}^1C_1\Delta\, f(x_0)}$$

$$f(x_0+2h)=\mathrm{E}^2\ f(x_0)$$
$$f(x_0+2h)=(1+\Delta)^2\ f(x_0)$$
$$f(x_0+2h)=\left[1+2\Delta+\Delta^2\right]f(x_0)$$
$$f(x_0+2h)=f(x_0)+2\,\Delta\, f(x_0)+\Delta^2\ f(x_0)$$

$$\boxed{f(x_0+2h)={}^2C_0\ f(x_0)+{}^2C_1\Delta\, f(x_0)+{}^2C_2\Delta^2\ f(x_0)}$$

Similarly, $f(x_0+3h)=\mathrm{E}^3\ f(x_0)$

$$f(x_0+3h)=(1+\Delta)^3\ f(x_0)$$
$$f(x_0+3h)=\left[1+3\Delta+3\Delta^2+\Delta^3\right]f(x_0)$$
$$f(x_0+3h)=f(x_0)+3\,\Delta\, f(x_0)+3\Delta^2\ f(x_0)+\Delta^3\ f(x_0)$$

$$\boxed{f(x_0+3h)={}^3C_0\ f(x_0)+{}^3C_1\Delta\, f(x_0)+{}^3C_2\Delta^2\ f(x_0)+{}^3C_3\Delta^3\ f(x_0)}$$

: :

: :

$$f(x_0+nh)=\mathrm{E}^n\ f(x_0)$$
$$f(x_0+nh)=(1+\Delta)^n\ f(x_0)$$
$$f(x_0+nh)=\left[{}^nC_0\,1^n\,\Delta^0+{}^nC_1\,1^{n-1}\,\Delta^1+{}^nC_2\,1^{n-2}\,\Delta^2+{}^nC_3\,1^{n-3}\,\Delta^3+.....+{}^nC_n\,1^{n-n}\,\Delta^n\right]f(x_0)$$
$$f(x_0+nh)=\left[{}^nC_0+{}^nC_1\ \Delta+{}^nC_2\ \Delta^2+{}^nC_3\ \Delta^3+.....+{}^nC_n\ \Delta^n\right]f(x_0)$$

$$\boxed{f(x_0+nh)={}^nC_0\ f(x_0)+{}^nC_1\ \Delta\, f(x_0)+{}^nC_2\ \Delta^2 f(x_0)+{}^nC_3\ \Delta^3 f(x_0)+.....+{}^nC_n\ \Delta^n\ f(x_0)}$$

Define Difference table and describe its uses ?

Let $'x'$ be the independent variable which takes values $x_0,\ x_1, x_2, x_3,.....,x_n$ with the interval of difference 'h'

Let y is a dependent variable which is expressed as the function of 'x' *i.e.*, $y=f(x)$

The functional values of y are $y_0=f(x_0),\ y_1=f(x_1),\ y_2=f(x_2),\ \ldots,\ y_n=f(x_n)$

Now the difference table constructed for known values of $y=f(x)$ at equal intervals by taking $\Delta y, \Delta^2 y, \Delta^3 y,.....\Delta^n y$ and $\nabla y, \nabla^2 y, \nabla^3 y,.....\nabla^n y$ in the columns.

Forward Difference tables is look likes below :

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$
x_0	y_0					
		$y_1-y_0=\Delta y_0$				
x_1	y_1		$\Delta y_1-\Delta y_0=\Delta^2 y_0$			
		$y_2-y_1=\Delta y_1$		$\Delta^2 y_1-\Delta^2 y_0=\Delta^3 y_0$		
x_2	y_2		$\Delta y_2-\Delta y_1=\Delta^2 y_1$		$\Delta^3 y_1-\Delta^3 y_0=\Delta^4 y_0$	
		$y_3-y_2=\Delta y_2$		$\Delta^2 y_2-\Delta^2 y_1=\Delta^3 y_1$		$\Delta^4 y_1-\Delta^4 y_0=\Delta^5 y_0$
x_3	y_3		$\Delta y_3-\Delta y_2=\Delta^2 y_2$		$\Delta^3 y_2-\Delta^3 y_1=\Delta^4 y_1$	
		$y_4-y_3=\Delta y_3$		$\Delta^2 y_3-\Delta^2 y_2=\Delta^3 y_2$		
x_4	y_4		$\Delta y_4-\Delta y_3=\Delta^2 y_3$			
		$y_5-y_4=\Delta y_4$				
x_5	y_5					

Backward Difference tables is look likes below :

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$	$\nabla^5 y$
x_0	y_0					
		$y_1-y_0=\nabla y_1$				
x_1	y_1		$\nabla y_2-\nabla y_1=\nabla^2 y_2$			
		$y_2-y_1=\nabla y_2$		$\nabla^2 y_3-\nabla^2 y_2=\nabla^3 y_3$		
x_2	y_2		$\nabla y_3-\nabla y_2=\nabla^2 y_3$		$\nabla^3 y_4-\nabla^3 y_3=\nabla^4 y_4$	
		$y_3-y_2=\nabla y_3$		$\nabla^2 y_4-\nabla^2 y_3=\nabla^3 y_4$		$\nabla^4 y_5-\nabla^4 y_4=\nabla^5 y_5$
x_3	y_3		$\nabla y_4-\nabla y_3=\nabla^2 y_4$		$\nabla^3 y_5-\nabla^3 y_4=\nabla^4 y_5$	
		$y_4-y_3=\nabla y_4$		$\nabla^2 y_5-\nabla^2 y_4=\nabla^3 y_5$		
x_4	y_4		$\nabla y_5-\nabla y_4=\nabla^2 y_5$			
		$y_5-y_4=\nabla y_5$				
x_5	y_5					

Theorem :

Statement : The nth order differences of nth degree polynomial are constants and $(n+1)^{th},(n+2)^{th},(n+3)^{th},.....$ order differences become zero.

Proof : Let $y = f(x) = a_0\ x^n + a_1\ x^{n-1} + a_2\ x^{n-2} + a_3\ x^{n-3} + ... + a_{n-1}\ x + a_n \rightarrow (1)$ be the nth degree polynomial.

If 'h' is the interval of difference then

$$f(x+h) = a_0\ (x+h)^n + a_1\ (x+h)^{n-1} + a_2\ (x+h)^{n-2} + a_3\ (x+h)^{n-3} + ... + a_{n-1}\ (x+h) + a_n \rightarrow (2)$$

By the definition of forward difference operator Δ

$$\Delta\ f(x) = f(x+h) - f(x)$$

$$\Delta\ f(x) = \left\{a_0\ (x+h)^n + a_1\ (x+h)^{n-1} + a_2\ (x+h)^{n-2} + a_3\ (x+h)^{n-3} + ... + a_{n-1}\ (x+h) + a_n\right\} - \left\{a_0\ x^n + a_1\ x^{n-1} + a_2\ x^{n-2} + a_3\ x^{n-3} + ... + a_{n-1}\ x + a_n\right\}$$

$$= a_0\left\{(x+h)^n - x^n\right\} + a_1\left\{(x+h)^{n-1} - x^{n-1}\right\} + a_2\left\{(x+h)^{n-2} - x^{n-2}\right\} + a_3\left\{(x+h)^{n-3} - x^{n-3}\right\} + ... + a_{n-1}\left\{x+h-x\right\} + a_n - a_n$$

$$\begin{aligned}
&= a_0\left\{{}^nC_0\ x^n\ h^0 + {}^nC_1\ x^{n-1}\ h^1 + {}^nC_2\ x^{n-2}\ h^2 + ... + {}^nC_{n-1}\ x^{n-(n-1)}\ h^{n-1} + {}^nC_n\ x^{n-n}\ h^n - x^n\right\} \\
&\quad + a_1\left\{{}^{n-1}C_0\ x^{n-1}\ h^0 + {}^{n-1}C_1\ x^{n-2}\ h^1 + {}^{n-1}C_2\ x^{n-3}\ h^2 + ... + {}^{n-1}C_{n-2}\ x^{(n-1)-(n-2)}\ h^{n-2} + {}^{n-1}C_{n-1}\ x^{(n-1)-(n-1)}\ h^{n-1} - x^{n-1}\right\} \\
&\quad + a_2\left\{{}^{n-2}C_0\ x^{n-2}\ h^0 + {}^{n-2}C_1\ x^{n-3}\ h^1 + {}^{n-2}C_2\ x^{n-4}\ h^2 + ... + {}^{n-2}C_{n-3}\ x^{(n-2)-(n-3)}\ h^{n-3} + {}^{n-2}C_{n-2}\ x^{(n-2)-(n-2)}\ h^{n-2} - x^{n-2}\right\} + ... + a_{n-1}h
\end{aligned}$$

$$= a_0\ nh\ x^{n-1} + x^{n-2}\left[a_0\ {}^nC_2h^2 + a_1\ {}^{n-1}C_1h\right] - x^{n-3}\left[a_0\ {}^nC_3h^3 + a_1\ {}^{n-1}C_2h^2 + a_2\ {}^{n-2}C_1h\right] + x^{n-4}\left[a_0\ {}^nC_4h^4 + a_1\ {}^{n-1}C_3h^3 + a_2\ {}^{n-2}C_2h^2 + a_2\ {}^{n-3}C_1h\right] + ... + a_{n-1}h$$

$$\Delta\ f(x) = a_0nh\ x^{n-1} + b_1x^{n-2} + b_2x^{n-3} + b_3x^{n-4} + ... + b_{n-2}x + b_{n-1} \rightarrow (3)$$

$$\Delta\ f(x+h) = a_0nh(x+h)^{n-1} + b_1(x+h)^{n-2} + b_2(x+h)^{n-3} + b_3(x+h)^{n-4} + ... + b_{n-2}(x+h) + b_{n-1}$$

By the definition $\Delta^2 f(x) = \Delta\ f(x+h) - \Delta\ f(x)$

$$\Delta^2\ f(x) = \left\{a_0nh(x+h)^{n-1} + b_1(x+h)^{n-2} + b_2(x+h)^{n-3} + b_3(x+h)^{n-4} + ... + b_{n-2}(x+h) + b_{n-1}\right\} - \left\{a_0nh\ x^{n-1} + b_1x^{n-2} + b_2x^{n-3} + b_3x^{n-4} + ... + b_{n-2}x + b_{n-1}\right\}$$

$$= a_0nh\left\{(x+h)^{n-1} - x^{n-1}\right\} + b_1\left\{(x+h)^{n-2} - x^{n-2}\right\} + b_2\left\{(x+h)^{n-3} - x^{n-3}\right\} + b_3\left\{(x+h)^{n-4} - x^{n-4}\right\} + ... + b_{n-2}\left\{(x+h) - x\right\} + b_{n-1} - b_{n-1}$$

$$= a_0 nh\left\{ {}^{n-1}C_0\ x^{n-1}\ h^0 + {}^{n-1}C_1\ x^{n-2}\ h^1 + {}^{n-1}C_2\ x^{n-3}\ h^2 + \ldots\ldots + {}^{n-1}C_{n-2}\ x^{(n-1)-(n-2)}\ h^{n-2} + {}^{n-1}C_{n-1}\ x^{(n-1)-(n-1)}\ h^{n-1} - x^{n-1} \right\}$$

$$+ b_1\left\{ {}^{n-2}C_0\ x^{n-2}\ h^0 + {}^{n-2}C_1\ x^{n-3}\ h^1 + {}^{n-2}C_2\ x^{n-4}\ h^2 + \ldots\ldots + {}^{n-2}C_{n-3}\ x^{(n-2)-(n-3)}\ h^{n-3} + {}^{n-2}C_{n-2}\ x^{(n-2)-(n-2)}\ h^{n-2} - x^{n-2} \right\}$$

$$+ b_2\left\{ {}^{n-3}C_0\ x^{n-3}\ h^0 + {}^{n-3}C_1\ x^{n-4}\ h^1 + {}^{n-3}C_2\ x^{n-5}\ h^2 + \ldots\ldots + {}^{n-3}C_{n-4}\ x^{(n-3)-(n-4)}\ h^{n-4} + {}^{n-3}C_{n-3}\ x^{(n-3)-(n-3)}\ h^{n-3} - x^{n-3} \right\} + \ldots\ldots\ldots + hb_{n-2}$$

$$= a_0\ n(n-1)h^2\ x^{n-2} + x^{n-3}\left[a_0 n\ {}^{n-1}C_2 h^3 + b_1\ {}^{n-2}C_1 h\right] + x^{n-4}\left[a_0 n\ {}^{n-1}C_3 h^4 + b_1\ {}^{n-2}C_2 h^2 + b_2\ {}^{n-2}C_1 h\right] + \ldots\ldots\ldots + hb_{n-2}$$

$$\Delta^2\ f(x) = a_0\ n(n-1)h^2\ x^{n-2} + c_1\ x^{n-3} + c_2\ x^{n-4} + c_3\ x^{n-5} + \ldots\ldots\ldots + c_{n-3}x + c_{n-2} \rightarrow (4)$$

Continue this process until we get n^{th} order differences, then

$$\Delta^n\ f(x) = a_0\ n(n-1)(n-2)(n-3)\ldots\ldots\ldots(n-(n-2))(n-(n-1))h^n\ x^{n-n}$$

$$\Delta^n\ f(x) = a_0\ n(n-1)(n-2)(n-3)\ldots\ldots\ldots(n-(n-2))(n-(n-1))h^n\ x^0$$

$$\Delta^n\ f(x) = a_0\ n(n-1)(n-2)(n-3)\ldots\ldots\ldots(n-(n-2))(n-(n-1))h^n$$

$$\Delta^n\ f(x) = a_0\ n!\,h^n$$

$\Delta^n\ f(x)$ is independent of 'x' *i.e.*, $\Delta^n\ f(x)$ is constant

n^{th} order difference of nth degree polynomial is constant.

$$\therefore \Delta^{n+1}\ f(x) = \Delta^n\ f(x+h) - \Delta^n\ f(x) = a_0\ n!\,h^n - a_0\ n!\,h^n$$

$\Delta^{n+1}\ f(x) = 0 \quad \therefore (n+1)^{th}$ order difference of nth degree polynomial is zero

$\therefore (n+1)^{th}, (n+2)^{th}, (n+3)^{th}, \ldots\ldots\ldots$ order differences of nth degree polynomial are zeros.

Problem 1 : Estimate Bengal population during the year 1911 from the following data

Year	1881	1891	1901	1911	1921	1931
Population	363	391	421	……	467	501

Solution : From the given data, let us consider x and y are the independent and dependent variables respectively.

i.e., x indicates the years and y indicates the population.

$\therefore$ The values of x and y are

$x_0 = 1881$, $x_1 = 1891$, $x_2 = 1901$, $x_3 = 1911$, $x_4 = 1921$, $x_5 = 1931$

$y_0 = 363$, $y_1 = 391$, $y_2 = 421$, $y_3 = a$, $y_4 = 467$, $y_5 = 501$

From the given data we know the 5 values of y

$\therefore$ 'y' is called as $n-1^{th}$ order polynomial in 'x'

i.e., $n-1 = 5-1 = 4$

i.e., 'y' is a 4th order polynomial in 'x'

We know that, n^{th} order difference of nth degree polynomial is constant and higher order differences become zeros.

i.e., $\Delta^5 y_0 = 0$

We know that, by the relationship between Δ and E

$$\Delta = \mathrm{E} - 1$$

$$\therefore \Delta^5 y_0 = (\mathrm{E}-1)^5 y_0 = 0$$

$$\left[{}^5C_0\mathrm{E}^5 - {}^5C_1\mathrm{E}^4 + {}^5C_2\mathrm{E}^3 - {}^5C_3\mathrm{E}^2 + {}^5C_4\mathrm{E} - {}^5C_5\mathrm{E}^0\right]y_0 = 0$$

$$\left[\mathrm{E}^5 - 5\mathrm{E}^4 + 10\mathrm{E}^3 - 10\mathrm{E}^2 + 5\mathrm{E} - 1\right]y_0 = 0$$

$$\mathrm{E}^5 y_0 - 5\mathrm{E}^4 y_0 + 10\mathrm{E}^3 y_0 - 10\mathrm{E}^2 y_0 + 5\mathrm{E}y_0 - y_0 = 0$$

$$y_5 - 5y_4 + 10y_3 - 10y_2 + 5y_1 - y_0 = 0$$

$$501 - 5(467) + 10a - 10(421) + 5(391) - 363 = 0$$

$$10a - 4452 = 0$$

$$a = \frac{4452}{10} = 445.2$$

$\therefore$ The estimated Bengal population in the year 1911 is 445.2

Problem 2 : Fill the following table

x	10	15	20	25	30	35
y	43	_	79	92	_	127

Solution : From the given data, let us consider x and y are the independent and dependent variables respectively.

$\therefore$ The values of x and y are

$x_0 = 10$, $x_1 = 15$, $x_2 = 20$, $x_3 = 25$, $x_4 = 30$, $x_5 = 35$

$y_0 = 43$, $y_1 = a$, $y_2 = 79$, $y_3 = 92$, $y_4 = b$, $y_5 = 127$

From the given data we know the 4 values of y

$\therefore$ 'y' is called as $n-1^{th}$ order polynomial in 'x' *i.e.*, $n-1 = 4-1 = 3$

i.e., 'y' is a 3rd order polynomial in 'x'

We know that, n^{th} order difference of nth degree polynomial is constant and higher order differences become zeros.

i.e., $\Delta^4 y_0 = 0$ $\Delta^5 y_0 = 0$

Using Forward Difference table

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$
10	43					
		$a-43$				
15	a		$122-2a$			
		$79-a$		$3a-188$		
20	79		$-66+a$		$b-4a+149$	
		13		$b-a-39$		$5a-5b+214$
25	92		$-105+b$		$a-4b+363$	
		$b-92$		$324-3b$		
30	b		$219-2b$			
		$127-b$				
35	127					

i.e., $\Delta^4 y_0 = 0$

i.e., $b-4a+149 = 0 \Rightarrow 4a-b = 149 \rightarrow (1)$

Similarly $\Delta^5 y_0 = 0$

i.e., $5a-5b+214 = 0 \Rightarrow 5a-5b = -214 \rightarrow (2)$

From Equation (i) and Equation (ii)

$$(1)\times 5 \Rightarrow 20a-5b = 745$$

$$(2)\times 1 \Rightarrow 5a-5b = -214$$

$$15a = 959$$

$$a = \frac{959}{15} = 63.9333 \cong 64$$

From Equation (1)

$$4(64)-b = 149 \Rightarrow 256-b = 149$$

$$b = 256-149$$

$$b = 107$$

$\therefore$ Missing values in the given data are $a = 64$ and $b = 107$

Problem 3 : Fill the following table

x	2	2.1	2.2	2.3	2.4	2.5	2.6
y	0.14	-	0.11	0.1	-	0.08	0.07

Solution : From the given data, let us consider x and y are the independent and dependent variables respectively.

$\therefore$ The values of x and y are

$x_0 = 2$, $x_1 = 2.1$, $x_2 = 2.2$, $x_3 = 2.3$, $x_4 = 2.4$, $x_5 = 2.5$ $x_6 = 2.6$

$y_0 = 0.14$, $y_1 = a$, $y_2 = 0.11$, $y_3 = 0.1$, $y_4 = b$, $y_5 = 0.08$ $y_6 = 0.07$

From the given data we know the 5 values of y

$\therefore$ 'y' is called as $n-1^{th}$ order polynomial in 'x' *i.e.*, $n-1=5-1=4$

i.e., 'y' is a 4rd order polynomial in 'x'

We know that, n^{th} order difference of nth degree polynomial is constant and higher order differences become zeros.

i.e., $\Delta^5 y_0 = 0$ $\Delta^6 y_0 = 0$

Using Forward Difference table

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$	$\Delta^5 y$	$\Delta^6 y$
2	0.14						
		$a-0.14$					
2.1	a		$0.25-2a$				
		$0.11-a$		$3a-0.37$			
2.2	0.11		$a-0.12$		$b-4a+0.4$		
		-0.01		$b-a+0.03$		$5a-5b-0.16$	
2.3	0.1		$b-0.09$		$a-4b+0.24$		$15b-6a-0.62$
		$b-0.1$		$0.27-3b$		$10b-a-0.78$	
2.4	b		$0.18-2b$		$6b-0.54$		
		$0.08-b$		$3b-0.27$			
2.5	0.08		$b-0.09$				
		-0.01					
2.6	0.07						

i.e., $\Delta^5 y_0 = 0$

i.e., $5a-5b-0.16=0 \Rightarrow 5a-5b=0.16 \rightarrow (1)$

Similarly $\Delta^6 y_0 = 0$

i.e., $15b-6a-0.62=0 \Rightarrow 15b-6a=0.62 \rightarrow (2)$

From Equation (i) and Equation (ii)

$(2)\times 1 \Rightarrow \quad 15b-6a=0.62$

$(1)\times 3 \Rightarrow \quad -15b+15a=0.48$

$9a \quad = 1.1$

$$a = \frac{1.1}{9} = 0.1222 \cong 0.12$$

From Equation (1)

$5(0.12)-5b=0.16 \Rightarrow 0.6-5b=0.16$

$5b = 0.6-0.16 = 0.44$

$$a = \frac{0.44}{5} = 0.088 \cong 0.09$$

$\therefore$ Missing values in the given data are $a = 0.12$ and $b = 0.09$

Problem 4: If $f(0)=-3$, $f(1)=6$, $f(2)=8$, $f(3)=12$ then find $f(5)$ values.

Solution : In the given data, the values of 'x' are 0, 1, 2, 3

And the values of 'y' are -3, 6, 8, 12

Here we know the 4 values of y

$\therefore$ y is a third order polynomial in 'x'

$i.e., y = ax^3 + bx^2 + cx + d$

Let $x=0$ then $y=d$

Given , $f(0)=-3$ $i.e.,$ $d=-3$

Let $x=1$ then $y=a+b+c-3=f(1)$

$\Rightarrow a+b+c-3=6$

$\Rightarrow a+b+c=9 \rightarrow (1)$

Let $x=2$ then $y=8a+4b+2c-3=f(2)$

$\Rightarrow 8a+4b+2c-3=8$

$\Rightarrow 8a+4b+2c=11 \rightarrow (2)$

Let $x=3$ then $y=27a+9b+3c-3=f(3)$

$\Rightarrow 27a+9b+3c-3=12$

$\Rightarrow 27a+9b+3c=15$ $\Rightarrow 9a+3b+c=5 \rightarrow (3)$

From (1) and (2)

$$\begin{aligned} 8a+4b+2c &= 11 \\ 2a+2b+2c &= 18 \\ \hline 6a+2b &= -7 \rightarrow (4) \end{aligned}$$

From (1) and (3)

$$\begin{aligned} 9a+3b+c &= 5 \\ a+\ \ b+c &= 9 \\ \hline 8a+2b &= -4 \rightarrow (5) \end{aligned}$$

From (4) and (5)

$$\begin{aligned} 8a+2b &= -4 \\ 6a+2b &= -7 \\ \hline 2a &= 3 \Rightarrow a = \frac{3}{2} \end{aligned}$$

From equation (4)

$$6\left(\frac{3}{2}\right)+2b=-7 \Rightarrow 2b=-7-9 \Rightarrow 2b=-16 \Rightarrow b=-\frac{16}{2}=-8$$

From equation (1) $\frac{3}{2}+(-8)+c=9 \Rightarrow c=9+8-\frac{3}{2} \Rightarrow c=\frac{31}{2}$

The constructed third order polynomial for the given data is $y=1.5x^3-8x^2+15.5x-3$

Put $x=5$ then $f(5)=1.5(5)^3-8(5)^2+15.5(5)-3=62$

Put $x=6$ then $f(6)=1.5(6)^3-8(6)^2+15.5(6)-3=126$

Problem 5 : If Δ is the forward difference operator then find $\Delta^n e^{ax+b}$

Solution : Consider, $f(x) = e^{ax+b}$

Let h be the interval of difference then $f(x+h) = e^{a(x+h)+b}$

By the definition of forward difference operator

$$\Delta f(x) = f(x+h) - f(x)$$

$$= e^{ax+ah+b} - e^{ax+b}$$

$$= e^{ax+b} e^{ah} - e^{ax+b}$$

$$= e^{ax+b}\left(e^{ah} - 1\right)$$

$$\Delta f(x+h) = e^{a(x+h)+b}\left(e^{ah} - 1\right) = e^{ax+b} e^{ah}\left(e^{ah} - 1\right)$$

Second order forward difference of $f(x)$ is

$$\Delta^2 f(x) = \Delta f(x+h) - \Delta f(x)$$

$$= e^{ax+b} e^{ah}\left(e^{ah} - 1\right) - e^{ax+b}\left(e^{ah} - 1\right)$$

$$= e^{ax+b}\left(e^{ah} - 1\right)\left(e^{ah} - 1\right)$$

$$\Delta^2 f(x) = e^{ax+b}\left(e^{ah} - 1\right)^2$$

Similarly, the nth order forward difference of $f(x)$ is

$$\Delta^n f(x) = e^{ax+b}\left(e^{ah} - 1\right)^n$$

Problem 6 : Assuming interval of difference as unity, find out $\left[\frac{\Delta^2}{\mathrm{E}}\right] x^3$ and $\frac{\Delta^2 x^3}{\mathrm{E}\, x^2}$

Solution : Given, interval of difference is unity *i.e.*, $h = 1$

By the relationship between Δ and E , $\Delta = \mathrm{E} - 1$

$$\left[\frac{\Delta^2}{\mathrm{E}}\right] x^3 = \frac{(1-\mathrm{E})^2}{\mathrm{E}} x^3 = \frac{\left(\mathrm{E}^2 - 2\mathrm{E} + 1\right)}{\mathrm{E}} x^3 = \left(\mathrm{E} - 2 + \mathrm{E}^{-1}\right) x^3 = \mathrm{E}\, x^3 - 2x^3 + \mathrm{E}^{-1} x^3$$

By the definition of shifting operator E

$$\mathrm{E}\, f(x) = f(x+h)$$

$$\mathrm{E}\, x^3 = (x+h)^3$$

$$\mathrm{E}\, x^3 = (x+1)^3$$

Similarly, $\mathrm{E}^{-1} x^3 = (x-h)^3 = (x-1)^3$

$$\left[\frac{\Delta^2}{\mathrm{E}}\right] x^3 = \mathrm{E}\, x^3 - 2x^3 + \mathrm{E}^{-1} x^3$$

$$\left[\frac{\Delta^2}{E}\right]x^3 = (x+1)^3 - 2x^3 + (x-1)^3$$

$$\left[\frac{\Delta^2}{E}\right]x^3 = \cancel{x^3} + \cancel{3x^2} + 3x + \cancel{1} - \cancel{2x^3} + \cancel{x^3} - \cancel{3x^2} + 3x - \cancel{1}$$

$$\left[\frac{\Delta^2}{E}\right]x^3 = 6x$$

(II) $\dfrac{\Delta^2 x^3}{E\,x^2} = \dfrac{(E-1)^2 x^3}{E\,x^2} = \dfrac{E^2 x^3 - 2E\,x^2 + x^3}{E\,x^2} = E\,x + E^{-1}x - 2x$

We know that $E\,f(x) = f(x+h)$

$$Ex = x + h = x + 1$$

Similarly, $E^{-1}x = x - 1$

$$\frac{\Delta^2 x^3}{E\,x^2} = E\,x + E^{-1}x - 2x = x + 1 + x - 1 - 2x = 0$$

Problem 7 :

S.T $u_0 + u_1 + u_2 + \ldots\ldots\ldots + u_n = {}^{(n+1)}C_1\, u_0 + {}^{(n+1)}C_2\, \Delta u_0 + {}^{(n+1)}C_3\, \Delta^2 u_0 + \ldots\ldots\ldots + {}^{(n+1)}C_{n+1}\, \Delta^n u_0$

Proof : By shifting operator E,

$$u_1 = E\,u_0 \quad u_2 = E^2\,u_0 \quad u_3 = E^3\,u_0 \quad \ldots\ldots \quad u_n = E^n\,u_0$$

$$\therefore\ u_0 + u_1 + u_2 + \ldots\ldots\ldots + u_n = u_0 + E\,u_0 + E^2\,u_0 + E^3\,u_0 + \ldots\ldots\ldots + E^n\,u_0$$

$$= u_0\left[1 + E + E^2 + E^3 + \ldots\ldots\ldots + E^n\right]$$

$$= u_0\left[\frac{1\left[E^{n+1} - 1\right]}{E - 1}\right] \qquad \left[In\ G.P, \quad S_n = \frac{a\left[r^n - 1\right]}{r - 1}\right]$$

We know that, $E = 1 + \Delta$

$$u_0 + u_1 + u_2 + \ldots\ldots\ldots + u_n = u_0\left[\frac{1\left[(1+\Delta)^{n+1} - 1\right]}{1 + \Delta - 1}\right] = \frac{1}{\Delta}\left[(1+\Delta)^{n+1} - 1\right]u_0$$

$$= \frac{1}{\Delta}\left[{}^{(n+1)}C_0 + {}^{(n+1)}C_1\,\Delta + {}^{(n+1)}C_2\,\Delta^2 + {}^{(n+1)}C_3\,\Delta^3 + \ldots\ldots + {}^{(n+1)}C_n\,\Delta^n + {}^{(n+1)}C_{n+1}\,\Delta^{n+1} - 1\right]u_0$$

$$= \frac{1}{\Delta}\left[\cancel{1} + {}^{(n+1)}C_1\,\Delta + {}^{(n+1)}C_2\,\Delta^2 + {}^{(n+1)}C_3\,\Delta^3 + \ldots\ldots + {}^{(n+1)}C_n\,\Delta^n + {}^{(n+1)}C_{n+1}\,\Delta^{n+1} - \cancel{1}\right]u_0$$

$$= \frac{1}{\Delta}\left[{}^{(n+1)}C_1\,\Delta + {}^{(n+1)}C_2\,\Delta^2 + {}^{(n+1)}C_3\,\Delta^3 + \ldots\ldots + {}^{(n+1)}C_n\,\Delta^n + {}^{(n+1)}C_{n+1}\,\Delta^{n+1}\right]u_0$$

$$= \left[{}^{(n+1)}C_1 + {}^{(n+1)}C_2\,\Delta + {}^{(n+1)}C_3\,\Delta^2 + \ldots\ldots + {}^{(n+1)}C_n\,\Delta^{n-1} + {}^{(n+1)}C_{n+1}\,\Delta^n\right]u_0$$

$$= {}^{(n+1)}C_1\,u_0 + {}^{(n+1)}C_2\,\Delta\,u_0 + {}^{(n+1)}C_3\,\Delta^2 u_0 + \ldots\ldots + {}^{(n+1)}C_n\,\Delta^{n-1}u_0 + {}^{(n+1)}C_{n+1}\,\Delta^n u_0$$

Problem 8 : ST

$$u_0+\frac{u_1}{1!}x+\frac{u_2}{2!}x^2+\frac{u_3}{3!}x^3+\dots\dots\dots=e^x\left\{u_0+\frac{x}{1!}\Delta u_0+\frac{x^2}{2!}\Delta^2u_0+\frac{x^3}{3!}\Delta^3u_0+\dots\dots\dots\dots\right\}$$

Proof : By the definition of shifting operator

$$\mathrm{E}\,u_0=u_1 \quad \mathrm{E}^2\,u_0=u_2 \quad \mathrm{E}^3\,u_0=u_3\ \dots\dots$$

Consider,

$$\begin{aligned}
u_0+\frac{u_1}{1!}x+\frac{u_2}{2!}x^2+\frac{u_3}{3!}x^3+\dots\dots\dots &= u_0+\frac{\mathrm{E}\,u_0}{1!}x+\frac{\mathrm{E}^2\,u_0}{2!}x^2+\frac{\mathrm{E}^3\,u_0}{3!}x^3+\dots\dots\dots\\
&= u_0\left(1+\frac{\mathrm{E}}{1!}x+\frac{\mathrm{E}^2}{2!}x^2+\frac{\mathrm{E}^3}{3!}x^3+\dots\dots\right)\\
&= u_0\left(1+\frac{\mathrm{E}\ x}{1!}+\frac{(\mathrm{E}\,x)^2}{2!}+\frac{(\mathrm{E}\,x)^3}{3!}+\dots\dots\right)\\
&= u_0\left(e^{\mathrm{E}\,x}\right)\\
&= u_0\left(e^{(1+\Delta)x}\right)\\
&= u_0\left(e^x e^{\Delta x}\right)\\
&= u_0\ e^x\left(1+\frac{\Delta x}{1!}+\frac{(\Delta x)^2}{2!}+\frac{(\Delta x)^3}{3!}+\dots\dots\dots\right)\\
&= u_0\ e^x\left(1+\frac{x}{1!}\Delta+\frac{x^2}{2!}\Delta^2+\frac{x^3}{3!}\Delta^3+\dots\dots\dots\right)
\end{aligned}$$

$$u_0+\frac{u_1}{1!}x+\frac{u_2}{2!}x^2+\frac{u_3}{3!}x^3+\dots\dots\dots=e^x\left\{u_0+\frac{x}{1!}\Delta u_0+\frac{x^2}{2!}\Delta^2u_0+\frac{x^3}{3!}\Delta^3u_0+\dots\dots\dots\dots\right\}$$

INTERPOLATION

Definition :

Generally the statistical data related to two mutually dependent variables in discrete series, now the method of estimating unknown values of dependent variable for some specified value of independent variable is called as interpolation.

If $x_0,\ x_0+h,\ x_0+2h,\ x_0+3h,.....x_0+nh$ are the values of independent variable then $y_0, y_1, y_2, y_3,.....y_n$ becomes the values of dependent variable respectively. Now, the method of estimating unknown values of 'y' for some specified values of x is called interpellation. Here the interpolated origin lies between minimum and maximum values of 'x'.

Example :

x	2	4	6	8	10	12
y	5	14	17	23	30	45

Be the values of a discrete series.

The method of estimating unknown values of $y = f(x)$ for some unknown values like $x = 3, 4.5, 5, 6.5, 7, 7.5,.5, 9, 9.5, 10.5, 11, 11.5$ is called interpolation.

Methods of interpolation:

1. If the arguments lies at equal intervals and the interpolated origin lies at the beginning of the given set of arguments then we will use Newton's forward formula for interpolation.
2. If the arguments lies at equal intervals and the interpolated origin lies at the end of the given set of arguments then we will use Newton's backward formula for interpolation.
3. If the arguments lies at equal intervals and the interpolated origin lies at the centre or middle of the given data then we will use different Central difference 26ormulae like Gauss forward and backward, Bessel's. Sterling's and Laplace Everett's formula for interpolation.
4. If the arguments lies at unequal intervals and the interpolated origin is either at the beginning or middle or end of the given set of arguments then we will use Lagrange's formula for interpolation.

Assumptions or conditions of interpolation:

1. Let $x_0,\ x_0+h,\ x_0+2h,\ x_0+3h,.....x_0+nh$ should have definite functional form.
2. The functional form $y = f(x)$ must be a nth degree polynomial.

3. The nth degree polynomial should have first and second order derivatives compulsory.
4. We should not apply numerical methods for trigonometric, exponential and logarithmic functions.
5. There should be any fluctuations in the data.
6. Sometimes trigonometric polynomials gives better estimate.
7. The function $y = f(x)$ can be shown either in polynomial form or in exponential series.

Uses of interpolation :

1. Generally Indian census will be conducted for every 10 year interval like 1951, 1961, 1971, 1981, 1991, 2001, 2011. Therefore we know the population size at the census year only. If you want to estimate the size of the population for middle of the Year then we will use different interpolation formula.
2. Interpolation can be used to determine some unknown polynomial relationship between x and y.

   For estimating future values of $y = f(x)$ we will use the methods of extrapolation.
3. By using different interpolation formula we can estimate first and second order derivatives.
4. By using numerical integration we will find the total of $y = f(x)$ at $x_0,\ x_0 + h,\ x_0 + 2h,\ x_0 + 3h, \ldots\ldots\ldots x_0 + nh$.

Newton's Forward Difference Formula :

Introduction :

This formula was introduced by two famous mathematician professor Newton and professor Gregory jointly. It takes forward differences from the difference tables for interpolation. Therefore it is called a Newton's forward interpolation formula.

Assumptions :

1. The function $y = f(x)$ must be a nth degree polynomial.
2. The arguments lies at equal intervals.
3. Interpolated origin lies at the beginning of the given set of arguments.
4. Here we must use forward difference operator Δ.

Statement : $x_0,\ x_0+h,\ x_0+2h,\ x_0+3h,.....x_0+nh$ be the values of independent variable lies at equal interval with the interval of difference 'h' then $y_0, y_1, y_2, y_3,.....y_n$ becomes the values of dependent variable $y = f(x)$.

If 'x' is the interpolated origin and x_0 is the beginning argument , 'h' is the interval of difference then the shift operator $u = \dfrac{x - x_0}{h}$

The Suitable Newton's forward interpolation formula for estimating unknown values of 'y' corresponding to known values of 'x' is given by

$$y = f(x) = y_0 + {}^uC_1 \Delta y_0 + {}^uC_2 \Delta^2 y_0 + {}^uC_3 \Delta^3 y_0 + ...+ {}^uC_n \Delta^n y_0$$

$$y = y_0 + u\Delta y_0 + \frac{u(u-1)}{2!}\Delta^2 y_0 + \frac{u(u-1)(u-2)}{3!}\Delta^3 y_0 + ... + \frac{u(u-1)(u-2).....(u-(n-1))}{n!}\Delta^n y_0$$

Proof :

Let x and y are 2 mutually dependent variables (x_0, y_0) ,(x_1, y_1) ,(x_2, y_2) ,(x_3, y_3) ,.....(x_n, y_n) becomes $n+1$ paired values of (x, y).

Now, the function $y = f(x)$ can be approximated as the nth degree polynomial passing through these $n+1$ paired values of (x, y)

$$y = f(x) = a_0 + a_1(x - x_0) + a_2(x - x_0)(x - x_1) + a_3(x - x_0)(x - x_1)(x - x_2) + ... + a_n(x - x_0)(x - x_1)(x - x_2).....(x - x_{n-1}) \rightarrow (1)$$

Generally , if we put $x = x_i$ then we get $y = y_i \ \forall \ i = 0, 1, 2, 3,.....$

Put $x = x_0$ in equation (1) then $y_o = a_0 \rightarrow (2)$

Put $x = x_1$ in equation (1) then

$$y_1 = a_0 + a_1(x_1 - x_0)$$

$$y_1 = a_0 + a_1 h$$

$$y_1 = y_0 + a_1 h \quad \Rightarrow \quad a_1 = \frac{y_1 - y_0}{h} \Rightarrow a_1 = \frac{\Delta y_0}{h} \rightarrow (3)$$

Similarly , if we put $x = x_2$, $x = x_3$, $x = x_4$, …… $x = x_n$ in equation (1) then we get

$$a_2 = \frac{\Delta^2 y_0}{h^2\, 2!} \rightarrow (4)$$

$$a_3 = \frac{\Delta^3 y_0}{h^3\, 3!} \rightarrow (5)$$

: :

$$a_n = \frac{\Delta^n y_0}{h^n\, n!} \rightarrow (n+2)$$

Substituting these $n+1$ equations [eq 2, eq 3, ……eq $n+2$] in equation (1) then we get

$$y = y_0 + \frac{\Delta y_0}{h1!}(x-x_0) + \frac{\Delta^2 y_0}{h^2\, 2!}(x-x_0)(x-x_1) + \frac{\Delta^3 y_0}{h^3\, 3!}(x-x_0)(x-x_1)(x-x_2) + \ldots\ldots$$
$$+ \frac{\Delta^n y_0}{h^n\, n!}(x-x_0)(x-x_1)(x-x_2)\ldots\ldots(x-x_{n-1}) \rightarrow (A)$$

By the shift operator $u = \dfrac{x-x_0}{h} \Rightarrow x - x_0 = uh \Rightarrow x = x_0 + uh$

Substituting these two values in equation(A), then

$$y = y_0 + \frac{\Delta y_0}{h1!}(uh) + \frac{\Delta^2 y_0}{h^2\, 2!}(uh)(x_0 + uh - x_1) + \frac{\Delta^3 y_0}{h^3\, 3!}(uh)(x_0 + uh - x_1)(x_0 + uh - x_2) + \ldots\ldots$$
$$+ \frac{\Delta^n y_0}{h^n\, n!}(uh)(x_0 + uh - x_1)(x_0 + uh - x_2)\ldots\ldots(x_0 + uh - x_{n-1})$$

$$y = y_0 + \frac{u}{1!}\Delta y_0 + \frac{\Delta^2 y_0}{h^2\, 2!}(uh)h(u-1) + \frac{\Delta^3 y_0}{h^3\, 3!}(uh)h(u-1)h(u-2) + \ldots\ldots$$
$$+ \frac{\Delta^n y_0}{h^n\, n!}(uh)h(u-1)h(u-2)\ldots\ldots h(u-(n-1))$$

$$y = y_0 + \frac{u}{1!}\Delta y_0 + \frac{\Delta^2 y_0}{h^2\, 2!}h^2 u(u-1) + \frac{\Delta^3 y_0}{h^3\, 3!}h^3 u(u-1)(u-2) + \ldots\ldots$$
$$+ \frac{\Delta^n y_0}{h^n\, n!}h^n u(u-1)(u-2)\ldots\ldots(u-(n-1))$$

$$y = y_0 + u\Delta y_0 + \frac{u(u-1)}{2!}\Delta^2 y_0 + \frac{u(u-1)(u-2)}{3!}\Delta^3 y_0 + \ldots\ldots + \frac{u(u-1)(u-2)\ldots\ldots(u-(n-1))}{n!}\Delta^n y_0$$

$$\boxed{y = f(x) = y_0 + {}^uC_1\Delta y_0 + {}^uC_2\Delta^2 y_0 + {}^uC_3\Delta^3 y_0 + \ldots\ldots + {}^uC_n\Delta^n y_0}$$

Problem 1 : Estimate the value of ‘ y ’ when $x = 2.8$ from the following data.

X	1	5	9	13	17	21
y	8	20	36	45	58	69

Solution : In the given data, difference between the argument of ‘ x ’ is uniform.

i.e., $h = x_1 - x_0 = 5 - 1 = 4$

The value to be estimated is lie at the starting of the given data. Hence, we have to use Newton’s Forward difference formula.

x	y	Δ	Δ^2	Δ^3	Δ^4	Δ^5
1	8					
		12				
5	20		4			
		16		-11		
9	36		-7		22	
		9		11		-39
13	45		4		-17	
		13		-6		
17	58		-2			
		11				
21	69					

From the given data, the interpolated origin $x = 2.8$

Here $x_0 = 1$, $h = 5 - 1 = 4$ shift operator $u = \dfrac{x - x_0}{h} = \dfrac{2.8 - 1}{4} = \dfrac{1.8}{4} = 0.45$

By Newton's forward difference formula

$$y = f(x) = y_0 + {}^uC_1 \Delta y_0 + {}^uC_2 \Delta^2 y_0 + {}^uC_3 \Delta^3 y_0 + {}^uC_4 \Delta^4 y_0 + {}^uC_5 \Delta^5 y_0$$

$$y = y_0 + u\Delta y_0 + \frac{u(u-1)}{2!}\Delta^2 y_0 + \frac{u(u-1)(u-2)}{3!}\Delta^3 y_0 + \frac{u(u-1)(u-2)(u-3)}{4!}\Delta^4 y_0 + \frac{u(u-1)(u-2)(u-3)(u-4)}{5!}\Delta^5 y_0$$

$$y = 8 + (0.45)12 + \frac{0.45(0.45-1)}{2} \times 4 + \frac{0.45(0.45-1)(0.45-2)}{6} \times (-11)$$
$$+ \frac{0.45(0.45-1)(0.45-2)(0.45-3)}{24} \times 22 + \frac{0.45(0.45-1)(0.45-2)(0.45-3)(0.45-4)}{120} \times (-39)$$

$$y = 8 + 5.4 - 0.4950 - 0.7033 - 0.8967 - 1.1286 = 10.1764 \cong 10$$

$\therefore$ The estimated value of y when $x = 2.8$ is 10.

Problem 2 : Estimate the number of students who got marks between 40 and 45 from the following data.

Marks	*40 – 50*	*50 – 60*	*60 – 70*	*70 – 80*	*80 -90*
No of Students	*45*	*30*	*15*	*20*	*29*

Solution : In the given data, contains classes and frequencies.

For this data, we have to calculate the cumulative frequencies then these frequencies become the value of ' y '.

The value to be estimated is lie at the starting of the given data. Hence, we have to use Newton's Forward difference formula.

Lower Bound	*Cumulative Frequencies*	Δ	Δ^2	Δ^3	Δ^4
40	*45*				
		30			
50	*75*		*-15*		
		15		*20*	
60	*90*		*5*		*-16*
		20		*4*	
70	*110*		*9*		
		29			
80	*139*				

From the given data, the interpolated origin $x = 45$

$$i.e., \quad h = x_1 - x_0 = 50 - 40 = 10$$

Here $x_0 = 40$, shift operator $u = \dfrac{x - x_0}{h} = \dfrac{45 - 40}{10} = 0.5$

By Newton's forward difference formula

$$y = f(x) = y_0 + {}^uC_1 \Delta y_0 + {}^uC_2 \Delta^2 y_0 + {}^uC_3 \Delta^3 y_0 + {}^uC_4 \Delta^4 y_0$$

$$y = y_0 + u\Delta y_0 + \frac{u(u-1)}{2!}\Delta^2 y_0 + \frac{u(u-1)(u-2)}{3!}\Delta^3 y_0 + \frac{u(u-1)(u-2)(u-3)}{4!}\Delta^4 y_0$$

$$y = 45 + (0.5)(30) + \frac{0.5(0.5-1)}{2} \times (-15) + \frac{0.5(0.5-1)(0.5-2)}{6} \times (20) + \frac{0.5(0.5-1)(0.5-2)(0.5-3)}{24} \times (-16)$$

$$y = 45 + 15 + 1.8750 + 1.25 + 0.6250 = 63.75 \cong 64$$

$\therefore$ From the data, the estimated number of students who got 45 marks is 64.

$\therefore$ Number of students who got marks between 40 and 45 is $= 64 - 45 = 19$

Newton's backward interpolation formula or Newton's backward formula :

Importance :

This formula was introduced by Two Famous mathematical scientists professor Newton and professor Gregory.

This formula takes backward difference from the difference table for interpolation. Therefore it is called as Newton's backward interpolation formula or Newton Gregory backward formula.

Assumptions or conditions :

1. The function $y = f(x)$ must be a nth degree polynomial.
2. The arguments or values of the independent variable must lie at equal intervals.

3. The interpolated origin should lie at ending or bottom of the given set of arguments.
4. Here, we must use the backward difference operator ∇.

State and prove Newton's backward difference formula :

Statement : If x and y are any two dependent variables such that $y = f(x)$ and x taking at equal interval of difference 'h' and the shift operator $u = \frac{x - x_n}{h}$ then

$$y = f(x) = y_n + {}^{u}C_1 \nabla y_n + {}^{(u+1)}C_2 \nabla^2 y_n + {}^{(u+2)}C_3 \nabla^3 y_n + {}^{(u+3)}C_4 \nabla^4 y_n + \ldots\ldots + {}^{(u+(n-1))}C_n \nabla^n y_n$$

$$y = f(x) = y_n + u\nabla y_n + \frac{(u+1)u}{2!}\nabla^2 y_n + \frac{(u+2)(u+1)u}{3!}\nabla^3 y_n + \frac{(u+3)(u+2)(u+1)u}{4!}\nabla^4 y_n + \ldots\ldots + \frac{(u+n-1)(u+n-2)(u+n-3)\ldots\ldots(u+2)(u+1)u}{n!}\nabla^n y_n$$

Proof :

Let x and y are 2 mutually dependent variables $(x_0, y_0), (x_1, y_1), (x_2, y_2), (x_3, y_3), \ldots\ldots(x_n, y_n)$ becomes $n+1$ paired values of (x, y).

Now, the function $y = f(x)$ can be approximated as the nth degree polynomial passing through these $n+1$ paired values of (x, y)

$$y = f(x) = a_0 + a_1(x - x_n) + a_2(x - x_n)(x - x_{n-1}) + a_3(x - x_n)(x - x_{n-1})(x - x_{n-2}) + \ldots\ldots\ldots\ldots + a_n(x - x_n)(x - x_{n-1})(x - x_{n-2})\ldots\ldots(x - x_0) \rightarrow (1)$$

Generally if $x = x_i$ the $y = y_i \;\; \forall \;\; i = 0, 1, 2, 3, \ldots\ldots n$

Put $x = x_n$ in equation (1) then $y_n = a_0 \rightarrow (2)$

Put $x = x_{n-1}$ in equation (1) then

$$y_{n-1} = a_0 + a_1(x_{n-1} - x_n)$$

$$y_{n-1} = y_n + a_1(-h)$$

$$a_1 h = y_n - y_{n-1}$$

$$a_1 = \frac{y_n - y_{n-1}}{h} \Rightarrow a_1 = \frac{\nabla y_n}{h \; 1!} \rightarrow (3)$$

Similarly, $$a_2 = \frac{\nabla^2 y_n}{h^2 \; 2!} \rightarrow (4)$$

$$a_3 = \frac{\nabla^3 y_n}{h^3 \; 3!} \rightarrow (5)$$

$$\vdots \qquad \vdots$$

$$a_n = \frac{\nabla^n y_n}{h^n \; n!} \rightarrow (n+2)$$

Substituting these $n+1$ equations [eq 2, eq 3, ……eq $n+2$] in equation (1) then we get

$$y = y_n + \frac{\nabla y_n}{h1!}(x - x_n) + \frac{\nabla^2 y_n}{h^2\, 2!}(x - x_n)(x - x_{n-1}) + \frac{\nabla^3 y_n}{h^3\, 3!}(x - x_n)(x - x_{n-1})(x - x_{n-2}) +$$

$$\ldots\ldots + \frac{\nabla^n y_n}{h^n\, n!}(x - x_n)(x - x_{n-1})(x - x_{n-2})\ldots\ldots(x - x_0)$$

By the shift operator $u = \dfrac{x - x_n}{h} \Rightarrow x - x_n = uh \Rightarrow x = x_n + uh$

$$y = y_n + \frac{\nabla y_n}{h1!}(uh) + \frac{\nabla^2 y_n}{h^2\, 2!}(uh)(x_n + uh - x_{n-1}) + \frac{\nabla^3 y_n}{h^3\, 3!}(uh)(x_n + uh - x_{n-1})(x_n + uh - x_{n-2}) +$$

$$\ldots\ldots + \frac{\nabla^n y_n}{h^n\, n!}(uh)(x_n + uh - x_{n-1})(x_n + uh - x_{n-2})\ldots\ldots(x_n + uh - x_0)$$

$$y = y_n + \frac{\nabla y_n}{h1!}(uh) + \frac{\nabla^2 y_n}{h^2\, 2!}(uh)(uh + h) + \frac{\nabla^3 y_n}{h^3\, 3!}(uh)(uh + h)(uh + 2h) +$$

$$\ldots\ldots + \frac{\nabla^n y_n}{h^n\, n!}(uh)(uh + h)(uh + 2h)\ldots\ldots(uh + (n-1)h)$$

$$y = y_n + \frac{\nabla y_n}{h1!}(uh) + \frac{\nabla^2 y_n}{h^2\, 2!}h^2 u\,(u+1) + \frac{\nabla^3 y_n}{h^3\, 3!}h^3 u\,(u+1)(u+2) +$$

$$\ldots\ldots + \frac{\nabla^n y_n}{h^n\, n!}h^n u\,(u+1)(u+2)\ldots\ldots(u + (n-1))$$

$$y = f(x) = y_n + u\nabla y_n + \frac{(u+1)u}{2!}\nabla^2 y_n + \frac{(u+2)(u+1)u}{3!}\nabla^3 y_n + \frac{(u+3)(u+2)(u+1)u}{4!}\nabla^4 y_n +$$

$$\ldots\ldots + \frac{(u+n-1)(u+n-2)(u+n-3)\ldots\ldots(u+2)(u+1)u}{n!}\nabla^n y_n$$

$$\boxed{y = f(x) = y_n + {}^{u}C_1\nabla y_n + {}^{(u+1)}C_2\nabla^2 y_n + {}^{(u+2)}C_3\nabla^3 y_n + {}^{(u+3)}C_4\nabla^4 y_n + \ldots\ldots + {}^{(u+(n-1))}C_n\nabla^n y_n}$$

Problem 1: Find the value of y when $x = 17.4$ from the following data.

X	*3*	*6*	*9*	*12*	*15*	*18*
y	*4*	*11*	*22*	*39*	*45*	*67*

Solution : In the given data, difference between the argument of 'x' is uniform.

i.e., $h = x_1 - x_0 = 6 - 3 = 3$

The value to be estimated is lie at the ending of the given data. Hence, we have to use Newton's Backward difference formula.

x	y	∇	∇^2	∇^3	∇^4	∇^5
3	4					
		7				
6	11		4			
		11		2		
9	22		6		-19	
		17		-17		63
12	39		-11		44	
		6		27		
15	45		16			
		22				
18	67					

From the given data, the interpolated origin $x = 17.4$

Here $x_n = 18$, $h = 6 - 3 = 3$ shift operator $u = \dfrac{x - x_n}{h} = \dfrac{17.4 - 18}{3} = \dfrac{-0.6}{3} = -0.2$

By Newton's backward difference formula

$$y = f(x) = y_n + u\nabla y_n + \frac{(u+1)u}{2!}\nabla^2 y_n + \frac{(u+2)(u+1)u}{3!}\nabla^3 y_n + \frac{(u+3)(u+2)(u+1)u}{4!}\nabla^4 y_n + \frac{(u+4)(u+3)(u+2)(u+1)(u)}{5!}\nabla^5 y_n$$

$$y = 67 + (-0.2)(22) + \frac{(-0.2+1)(-0.2)}{2}(16) + \frac{(-0.2+2)(-0.2+1)(-0.2)}{6}(27) + \frac{(-0.2+3)(-0.2+2)(-0.2+1)(-0.2)}{24}(44) + \frac{(-0.2+4)(-0.2+3)(-0.2+2)(-0.2+1)(-0.2)}{120}(63)$$

$$y = 67 - 4.4 - 1.28 - 12.96 - 1.4784 - 1.6088 = 56.9368 \cong 57$$

$\therefore$ Value of y when $x = 17.4$ is 57

Problem 2: Estimate the population during 1945 and 1975 from the following data.

Year	*1941*	*1951*	*1961*	*1971*	*1981*
Population	*46*	*60*	*81*	*93*	*101*

Solution : In the given data, difference between the argument of 'x' is uniform.

Here 1945 is lie at the starting of the given data , hence we have to use Newton's forward difference formula.

X	y	Δ	Δ^2	Δ^3	Δ^4
1941	46				
		14			
1951	60		7		
		21		-16	
1961	81		-9		21
		12		5	
1971	93		-4		
		8			
1981	101				

From the given data, the interpolated origin $x = 1945$

$$i.e., \quad h = x_1 - x_0 = 1951 - 1941 = 10$$

Here $x_0 = 1941$, shift operator $u = \dfrac{x - x_0}{h} = \dfrac{1945 - 1941}{10} = 0.4$

By Newton's forward difference formula

$$y = f(x) = y_0 + {}^uC_1 \Delta y_0 + {}^uC_2 \Delta^2 y_0 + {}^uC_3 \Delta^3 y_0 + {}^uC_4 \Delta^4 y_0$$

$$y = y_0 + u\Delta y_0 + \frac{u(u-1)}{2!}\Delta^2 y_0 + \frac{u(u-1)(u-2)}{3!}\Delta^3 y_0 + \frac{u(u-1)(u-2)(u-3)}{4!}\Delta^4 y_0$$

$$y = 46 + (0.4)(14) + \frac{0.4(0.4-1)}{2} \times (7) + \frac{0.4(0.4-1)(0.4-2)}{6} \times (-16) + \frac{0.4(0.4-1)(0.4-2)(0.4-3)}{24} \times (21)$$

$$y = 46 + 5.6 - 0.84 - 1.024 - 0.8736 = 48.8624 \cong 49$$

$\therefore$ From the data, the estimated population in the year 1945 is 49.

The value to be estimated is lie at the ending of the given data. Hence, we have to use Newton's Backward difference formula.

x	y	∇	∇^2	∇^3	∇^4
1941	46				
		14			
1951	60		7		
		21		-16	
1961	81		-9		21
		12		5	
1971	93		-4		
		8			
1981	101				

From the given data, the interpolated origin $x = 1975$

Here $x_n = 1981$, $h = 1951 - 1941 = 10$

shift operator $u = \frac{x - x_n}{h} = \frac{1975 - 1981}{10} = \frac{-6}{10} = -0.6$

By Newton's backward difference formula

$$y = f(x) = y_n + u\nabla y_n + \frac{(u+1)u}{2!}\nabla^2 y_n + \frac{(u+2)(u+1)u}{3!}\nabla^3 y_n + \frac{(u+3)(u+2)(u+1)u}{4!}\nabla^4 y_n$$

$$y = 101 + (-0.6)(8) + \frac{(-0.6+1)(-0.6)}{2}(-4) + \frac{(-0.6+2)(-0.6+1)(-0.6)}{6}(5) + \frac{(-0.6+3)(-0.6+2)(-0.6+1)(-0.6)}{24}(21)$$

$$y = 101 - 4.8 + 0.48 - 0.28 - 0.7056 = 95.6944 \cong 96$$

$\therefore$ Here the estimated population in the year 1975 is 96

Interpolation at unequal intervals :

If the values of the independent variable lie at equal intervals then we will use Newton's forward difference formula and Newton's backward difference formula for interpolation.

Sometimes the arguments may not be lie at equal intervals in such situations we have to use Newton's divided difference formula and Lagrange's formula for interpolation.

If the arguments life at unequal intervals then we have to to use Newton's divided difference formula and Lagrange's formula for interpolation

Lagrange's Formula :

Introduction : This formula was introduced by professor Lagrange. If the arguments lies at equal or unequal intervals then we will use this for interpolation.

Conditions or assumptions :

1. The function $y = f(x)$ must become a nth degree polynomial.
2. The argument may lie weather at equal or unequal intervals.
3. The interpolated origin lies at anywhere within the minimum and maximum values of given set of arguments.
4. Here we do not construct any difference table.

Uses :

1. Lagrange's formula can be used to estimate unknown value of $y = f(x)$ directly without constructing any difference table.
2. It can be used to determine some unknown polynomial relationship between x and $y = f(x)$ values.
3. When comparing with Newton's forward formula, Newton's backward formula and other Central difference formula the derivation of Lagrange's formula become easy for interpolation.

Merits and demerits :

1. Lagrange's formula becomes easy to understand.
2. It becomes easy to calculate.
3. There is no necessity for constructing the difference table.
4. We can directly interpolate, extrapolate unknown value of y for known value of x
5. When comparing with Newton's divided difference formula, the calculations of Lagrange's formula becomes difficult .
6. The application of this formula takes a lot of time.
7. While calculating Lagrange's formula we get some positive and negative signs in the numerator and denominator for this formula.
8. Therefore, there will be some chance of error that will enter into the calculations.

Statement :

If $x_0,\ x_1,\ x_2,\ x_3,.....x_n$ are the values of independent variable which lies at equal or unequal intervals then $y_0,\ y_1,\ y_2,\ y_3,.....y_n$, become the values of the dependent variable.

Now, the suitable Lagrange's interpolation formula for estimating unknown values of y corresponding to the known values of x is given by

$$f(x)=\frac{(x-x_1)(x-x_2)(x-x_3).....(x-x_n)}{(x_0-x_1)(x_0-x_2)(x_0-x_3).....(x_0-x_n)}y_0+\frac{(x-x_0)(x-x_2)(x-x_3).....(x-x_n)}{(x_1-x_0)(x_1-x_2)(x_1-x_3).....(x_1-x_n)}y_1$$
$$+\frac{(x-x_0)(x-x_1)(x-x_3).....(x-x_n)}{(x_2-x_0)(x_2-x_1)(x_2-x_3).....(x_2-x_n)}y_2+\frac{(x-x_0)(x-x_1)(x-x_2).....(x-x_n)}{(x_3-x_0)(x_3-x_1)(x_3-x_2).....(x_3-x_n)}y_3$$
$$+.....+\frac{(x-x_0)(x-x_1)(x-x_2).....(x-x_{n-1})}{(x_n-x_0)(x_n-x_1)(x_n-x_2).....(x_n-x_{n-1})}y_n$$

Proof : Let x and y are any 2 mutually dependent variables,

$(x_0,y_0)(x_1,y_1)(x_2,y_2),.....,(x_n,y_n)$ becomes paired values of (x,y)

Now, the function $y=f(x)$ can be approximated as the nth degree polynomial through the above $(n+1)$ paired values of (x,y)

$$y=f(x)=a_0(x-x_1)(x-x_2)(x-x_3).....(x-x_n)+a_1(x-x_0)(x-x_2)(x-x_3).....(x-x_n)$$
$$+a_2(x-x_0)(x-x_1)(x-x_3).....(x-x_n)+a_3(x-x_0)(x-x_1)(x-x_2)(x-x_4).....(x-x_n)$$
$$+.....+a_n(x-x_0)(x-x_1)(x-x_2).....(x-x_{n-1})\rightarrow(1)$$

Let equation (1) can be considered as the nth degree polynomial with respect to (x,y) where $x=x_i,\ y=y_i$.

Put $x=x_0$ in equation (1)

$$y_0=a_0(x_0-x_1)(x_0-x_2)(x_0-x_3).....(x_0-x_n)$$

$$a_0=\frac{y_0}{(x_0-x_1)(x_0-x_2)(x_0-x_3).....(x_0-x_n)}\rightarrow(2)$$

Put $x=x_1$ in equation (1)

$$y_1=a_1(x_1-x_0)(x_1-x_2)(x_1-x_3).....(x_1-x_n)$$

$$a_1=\frac{y_1}{(x_1-x_0)(x_1-x_2)(x_1-x_3).....(x_1-x_n)}\rightarrow(3)$$

Similarly $a_2 = \dfrac{y_2}{(x_2 - x_0)(x_2 - x_1)(x_2 - x_3).....(x_2 - x_n)} \rightarrow (4)$

$$a_3 = \frac{y_3}{(x_3 - x_0)(x_3 - x_1)(x_3 - x_2).....(x_3 - x_n)} \rightarrow (5)$$

$$\vdots \qquad \vdots$$

$$a_n = \frac{y_n}{(x_n - x_0)(x_n - x_1)(x_n - x_2).....(x_n - x_{n-1})} \rightarrow (n+1)$$

Substitute equation (2) , (3) , (4), ... (n+1) in equation (1) then we get

$$y = f(x) = \frac{(x - x_1)(x - x_2)(x - x_3).....(x - x_n)}{(x_0 - x_1)(x_0 - x_2)(x_0 - x_3).....(x_0 - x_n)} y_0 + \frac{(x - x_0)(x - x_2)(x - x_3).....(x - x_n)}{(x_1 - x_0)(x_1 - x_2)(x_1 - x_3).....(x_1 - x_n)} y_1$$

$$+ \frac{(x - x_0)(x - x_1)(x - x_3).....(x - x_n)}{(x_2 - x_0)(x_2 - x_1)(x_2 - x_3).....(x_2 - x_n)} y_2 + \frac{(x - x_0)(x - x_1)(x - x_2).....(x - x_n)}{(x_3 - x_0)(x_3 - x_1)(x_3 - x_2).....(x_3 - x_n)} y_3$$

$$+ ..... + \frac{(x - x_0)(x - x_1)(x - x_2).....(x - x_{n-1})}{(x_n - x_0)(x_n - x_1)(x_n - x_2).....(x_n - x_{n-1})} y_n$$

Problem : Find the value of y from the following data when $x = 22$ using Lagrange's formula.

X	*5*	*6*	*9*	*13*	*19*	*28*
y	*10*	*13*	*21*	*46*	*89*	*126*

Solution : Here the value of $x = 22$

By Lagrange's formula

$$y = f(x) = \frac{(x - x_1)(x - x_2)(x - x_3)(x - x_4)(x - x_5)}{(x_0 - x_1)(x_0 - x_2)(x_0 - x_3)(x_0 - x_4)(x_0 - x_5)} y_0 + \frac{(x - x_0)(x - x_2)(x - x_3)(x - x_4)(x - x_5)}{(x_1 - x_0)(x_1 - x_2)(x_1 - x_3)(x_1 - x_4)(x_1 - x_5)} y_1$$

$$+ \frac{(x - x_0)(x - x_1)(x - x_3)(x - x_4)(x - x_5)}{(x_2 - x_0)(x_2 - x_1)(x_2 - x_3)(x_2 - x_4)(x_2 - x_5)} y_2 + \frac{(x - x_0)(x - x_1)(x - x_2)(x - x_4)(x - x_5)}{(x_3 - x_0)(x_3 - x_1)(x_3 - x_2)(x_3 - x_4)(x_3 - x_5)} y_3$$

$$+ \frac{(x - x_0)(x - x_1)(x - x_2)(x - x_3)(x - x_5)}{(x_4 - x_0)(x_4 - x_1)(x_4 - x_2)(x_4 - x_3)(x_4 - x_5)} y_4 + \frac{(x - x_0)(x - x_1)(x - x_2)(x - x_3)(x - x_4)}{(x_5 - x_0)(x_5 - x_1)(x_5 - x_2)(x_5 - x_3)(x_5 - x_4)} y_5$$

$$= \frac{(22-6)(22-9)(22-13)(22-19)(22-28)}{(5-6)(5-9)(5-13)(5-19)(5-28)} \times 10 + \frac{(22-5)(22-9)(22-13)(22-19)(22-28)}{(6-5)(6-9)(6-13)(6-19)(6-28)} \times 13$$

$$+ \frac{(22-5)(22-6)(22-13)(22-19)(22-28)}{(9-5)(9-6)(9-13)(9-19)(9-28)} \times 21 + \frac{(22-5)(22-6)(22-9)(22-19)(22-28)}{(13-5)(13-6)(13-9)(13-19)(13-28)} \times 46$$

$$+ \frac{(22-5)(22-6)(22-9)(22-13)(22-28)}{(19-5)(19-6)(19-9)(19-13)(19-28)} \times 89 + \frac{(22-5)(22-6)(22-9)(22-13)(22-19)}{(28-5)(28-6)(28-9)(28-13)(28-19)} \times 126$$

$$= \frac{16 \times 13 \times 9 \times 3 \times (-6)}{(-1)(-4)(-8)(-14)(-23)} \times 10 + \frac{17 \times 13 \times 9 \times 3 \times (-6)}{1 \times (-3)(-7)(-13)(-22)} \times 13 + \frac{17 \times 16 \times 9 \times 3 \times (-6)}{4 \times 3 \times (-4)(-10)(-19)} \times 21$$

$$+ \frac{17 \times 16 \times 13 \times 3 \times (-6)}{8 \times 7 \times 4 \times (-6)(-15)} \times 46 + \frac{17 \times 16 \times 13 \times 9 \times (-6)}{14 \times 13 \times 10 \times 6 \times (-9)} \times 89 + \frac{17 \times 16 \times 13 \times 9 \times 3}{23 \times 22 \times 19 \times 15 \times 9} \times 126$$

$$= 32.7019 - 77.4935 + 101.4632 - 145.2286 + 172.9143 + 9.2685 = 93.6258 \cong 94$$